

MATHEMATICS Class XII - 2014

(English Version)

Group – A

1. Answer the following questions. (alternatives are to be noted): (1X10 = 10)
- (a) Two dice are thrown simultaneously. The probability of sum 4 of two numbers so obtained is:
- (i) $\frac{1}{12}$
 - (ii) $\frac{1}{36}$
 - (iii) $\frac{1}{18}$
 - (iv) $\frac{1}{9}$
- (b) The condition for the expression of $(2 - 3x)^{-\frac{1}{4}}$ is
- (i) $|x| > \frac{2}{3}$
 - (ii) $|x| \geq \frac{3}{2}$
 - (iii) $|x| < \frac{2}{3}$
 - (iv) $|x| \leq \frac{3}{2}$
- OR**
- The value of $\log_3 \left(\frac{1}{2} - \frac{1}{4} + \frac{1}{8} - \frac{1}{16} + \dots \right)$ is
- (i) 0
 - (ii) 1
 - (iii) -1
 - (iv) $\frac{1}{3}$
- (c) Which of the following is the equation of the parabola with vertex at the origin and directrix $y = 2$?
- (i) $y^2 = 8x$
 - (ii) $y^2 = -8x$
 - (iii) $x^2 = 8y$
 - (iv) $x^2 = -8y$

OR

Fill in the blanks:

The eccentricity of the hyperbola $9x^2 - 4y^2 + 36 = 0$ is

(d) Fill in the blanks:

If $f(x) = e^x$, $g(x) = 2 \log_e x$ and $F(x) = f\{g(x)\}$, then $\frac{dF}{dx} = \dots\dots\dots$

OR

Which of the following is the value of $\frac{d}{dx} \{ |x-1| + |x-5| \}$ at the point $x = 3$?

- (i) -2
- (ii) 0
- (iii) 2
- (iv) 4

(e) Which of the following will be correct if $y = p \sin(\alpha x) + q \cos(\alpha x)$?

- (i) $y_2 + \alpha^2 y = 0$
- (ii) $y_2 - \alpha y = 0$
- (iii) $y_2 + \alpha y = 0$
- (iv) $y_2 - \alpha^2 y = 0$

 $(y_1, y_2 \text{ have their usual meaning; } p, q, \alpha \text{ are constants.})$

(f) Fill in the blanks:

The value of $\int_{-1}^1 (1 + x + 3x^3 + 5x^5 + \dots + 99x^{99}) dx$ is

OR

Fill in the blanks:

If $\int f(x) dx = \frac{e^x}{2} (\sin x - \cos x)$, then $f(x)$ will be(g) Which of the following relations is satisfied by the function $f(x) = \int_1^x \frac{dt}{t}$?

- (i) $f(x+y) = f(x) + f(y)$
- (ii) $f\left(\frac{x}{y}\right) = f(x) - f(y)$
- (iii) $f(xy) = f(x)f(y)$
- (iv) $f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}$

- (h) Order and degree of the differential equation $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^{\frac{1}{3}} + x^{\frac{1}{4}} = 0$ are respectively:
- (i) 2 and 3
 - (ii) 2 and 12
 - (iii) 2 and 6
 - (iv) 2 and 4
- (i) A particle moves according to the law $s = t^3 - 9t^2 + 24t$. The distance covered by the particle before it first comes to rest is:
- (i) 10 unit
 - (ii) 16 unit
 - (iii) 20 unit
 - (iv) 24 unit
- (j) State the following statement is *True* or *False*:
 “ $f(x) = |x|$ has no minimum value.”

Group – B

2.

- (a) Answer any two questions: (2X2 = 4)
- (i) Without expanding show that “

$$\begin{vmatrix} a & b & c \\ x & y & z \\ yz & zx & xy \end{vmatrix} = \begin{vmatrix} ax & by & cz \\ x^2 & y^2 & z^2 \\ 1 & 1 & 1 \end{vmatrix}$$
 - (ii) Two dice are thrown simultaneously. What is the probability of getting an even number at the first die and the total of the two numbers so obtained is 8?
 - (iii) If the matrix A , $A = A^{-1}$, then show that $A(A^3 + I) = A + I$. (I is the unit matrix.)
- (b) Answer any one questions: (2X1 = 2)
- (i) What is the eccentricity of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, if length of its minor axis is equal to the distance between its foci?
 - (ii) Find the vertex of the parabola $y = -2x^2 + 12x - 17$.
- (c) Answer any one question: (2X1 = 2)
- (i) If $y = x + \frac{x^3}{3} + \frac{x^5}{5} + \dots + \infty$, show that $\frac{dy}{dx} = \frac{1}{1-x^2}$.
 - (ii) If $y = a \cos(\log x) + b \sin(\log x)$, show that $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = 0$, where a and b are constants.

(d) Answer any one question: (2X1 = 2)

(i) If f is integrable function in the interval $[-a, a]$ then show that:

$$\int_{-a}^a f(x) dx = \int_0^a [f(x) + f(-x)] dx.$$

(ii) Using the definition of definite integral as the limit of a sum evaluate $\int_0^2 ax dx$, where a is a constants.

(e) Answer any one question: (2X1 = 2)

(i) Find the differential equation of all circles passing through the origin and having their centres on the x-axis.

(ii) Find the equation of the curve which passes through the point (4, 3) and the gradient of the tangent to the curve at any point on it equal to the reciprocal of the ordinate of the point.

(f) Answer any three questions: (2X3 = 6)

(i) If $y = a \log|x| + bx^2 + x$ has extreme values at $x = -1$ and $x = 2$, find the values of a and b .

(ii) Find the area of the region bounded by the curve $y = (x-1)(5-x)$ and x-axis.

(iii) Find the values of x for which the function $f(x) = x^3 - 7x^2 + 8x - 10$ is monotonic increasing.

(iv) The tangent at a point (a, b) to the curve $y = \sin x$ is parallel to the line $2y = x$. If $a > 0$, find b .

(v) A particle moves in a straight line and its velocity v at a time t seconds is given by $v = (6t^2 - 2t + 3) \text{ cm/sec}$. Find the distance travelled by the particle during the first 5 seconds after the start.

Group – C

3.

(a) Answer any two questions: (4X2 = 8)

(i) Show that the matrix $aA = \begin{bmatrix} 2 & -3 \\ 3 & 4 \end{bmatrix}$ satisfies the equation $A^2 - 6A + 17I = 0$ and

hence find A^{-1} where I is the identity matrix and 0 is the null matrix of order 2×2 .

(ii) Using the principle of mathematical induction prove that $3^{2n} + 7$ is always divisible by 8 for all possible integers n .

(iii) Among two boxes, the first one contains 3 black and 5 white balls and the second one contains 5 black and 3 white balls. Two balls of same colour are transferred from the first box to the second, after which one ball is drawn at random from the second box. What is the probability that it is black?

(b) Answer any two questions:

(4X2 = 8)

- (i) The eccentricity of an ellipse is $\frac{1}{\sqrt{3}}$, the co-ordinate of focus is $(-2, 1)$ and the point of intersection of the major axis and the directrix is $(-2, 3)$. Find the co-ordinates of the centre of the ellipse and also equation to the ellipse.
- (ii) The slope of a chord of the hyperbola $y^2 = 4x$ is 2. Show that the locus of the point which divides the chord internally in the ratio 1:2 is a parabola whose equation is $\left(y - \frac{8}{9}\right)^2 = \frac{4}{9}\left(x - \frac{2}{9}\right)$.
- (iii) If foci of a hyperbola are $(\pm 6, 3)$ and eccentricity is $\frac{3}{2}$, find the length of the latus rectum.

(c) Answer any two questions:

(4X2 = 8)

- (i) Find $\int \left\{ \log(\log x) + \frac{1}{(\log x)^2} \right\} dx$.
- (ii) Find the value of $\lim_{n \rightarrow \infty} \left[\frac{1}{\sqrt{n^2 - 1^2}} + \frac{1}{\sqrt{n^2 - 2^2}} + \frac{1}{\sqrt{n^2 - 3^2}} + \dots + \frac{1}{\sqrt{n^2 - (n-1)^2}} \right]$.
- (iii) If $f(x) = (|x-2| + |x-4|)$, then evaluate $\int_1^4 f(x) dx$.

(d) Answer any two questions:

(4X2 = 8)

- (i) Solve $(e^y + 1) \cos x dx + \sin y dy = 0$.

(ii) solve $\frac{dy}{dx} = \frac{y \left(x \cos \frac{y}{x} + y \sin \frac{y}{x} \right)}{x \left(y \sin \frac{y}{x} - x \cos \frac{y}{x} \right)}$.

- (iii) Determine the particular solution of:

$$\sin^{-1} \left(\frac{d^2 y}{dx^2} - 1 \right) = x, \quad \frac{dy}{dx} = 0, \quad y = 0, \quad \text{at } x = 0.$$

(e) Answer any four questions:

(4X4 = 16)

- (i) If $y = mx + c$ is the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at a point on it, using the calculus show that $c^2 = a^2 m^2 + b^2$. Find the co-ordinates of the point of contact.
- (ii) Using calculus find the area of the region bounded by $y^2 = 8x$ and $y = x$ (a rough sketch is necessary).

- (iii) A spherical ice ball melts in such a way that the rate of melting is proportional to its volume of ice at that instant. If half the quantity of ice melts in 30 minutes, show that after 90 minutes from the start of melting, the volume of ice that remains is $\frac{1}{8}$ th of the original volume of the ice.
- (iv) Two particles move in a straight line. Particle A starts at a speed of 35 m/sec with a uniform acceleration 2 m/sec^2 . After 5 seconds particle B starts from the same point in the same direction with a uniform speed of 65 m/sec. Find the minimum distance between the particles.
- (v) A particle moves along a straight line. Its distance from a fixed point O on the line at a time t is $x = a \cos(nt + \theta)$, where a, n, θ are constants. Show that at time t its velocity v is given by $v^2 = n^2(a^2 - x^2)$. Find out the square of the acceleration of the particle when its velocity is zero.
- (vi) After leaving the ground from rest a balloon is rising vertically with uniform acceleration of 4 ft/sec^2 . After 16 seconds a small heavy particles fall freely from the balloon. When the particle will reach the ground? (Apply calculus).
- (vii) Using the calculus show that the portion of the normal to the curve $y = \frac{a}{2} \left(e^{\frac{x}{a}} + e^{-\frac{x}{a}} \right)$

at (x_1, y_1) intercepted between the curve and the x-axis is $\frac{y_1^2}{a}$.

Group – D

4. Answer the following questions as per the instructions. (6X4 = 24)

{Answer any one question from (a) and (b).}
 {Answer any one question from (c) and (d).}
 {Answer any two question from (e), (f), (g) and (h).}

(a)

- (i) Show that the determinant $\begin{vmatrix} 1 & a & a^2 \\ a^2 & 1 & a \\ a & a^2 & 1 \end{vmatrix}$ is a perfect square.

- (ii) Find the coefficient of x^{10} in the expansion of $\frac{1+x+x^2}{e^x}$.

(b)

- (i) If $A = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, show that $AA^T = I_3$.

- (ii) Find the sum of the infinite series $1 + \frac{3}{1!} + \frac{5}{2!} + \frac{7}{3!} + \dots + \infty$.
- (c)
- (i) If $f(x) = \begin{cases} x + \sin x & x < 0 \\ 0 & x \geq 0 \end{cases}$, examine whether $f'(x)$ is continuous at $x = 0$.
- (ii) If $y = \left(\frac{1+x}{1-x}\right)^n$, prove that $(1-x^2) \frac{d^2y}{dx^2} = 2(n+x) \frac{dy}{dx}$.
- (d)
- (i) If $x = \log(1+t^2)$ and $y = t - \tan^{-1} t$, find $\frac{dy}{dx}$.
- (ii) If $x = \sin \theta$ and $y = \cos p\theta$, where p is a constant, prove that:
 $(1-x^2)y_2 - xy_1 + p^2y = 0$.
- (e)
- (i) Find $\int \frac{dx}{3\cos x - 4\sin x + 5}$.
- (ii) Find $\int_1^3 \frac{\sqrt{x+1}}{\sqrt{x+1} + \sqrt{5-x}} dx$.
- (f)
- (i) Find $\int e^x [\log(\sec x + \tan x) + \sec x] dx$.
- (ii) Show that $\int_0^1 \sqrt{(x-1)(2-x)} dx = \frac{\pi}{8}$.
- (g)
- (i) Find $\int \frac{3x+1}{\sqrt{2-3x-2x^2}} dx$.
- (ii) Prove that $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$.
- (h)
- (i) Find $\int \frac{\log x}{(1+\log x)^2} dx$.
- (ii) Show that $\int_0^1 (\cos^{-1} x)^2 dx = \pi - 2$.
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